

Let's assume we adjust our hedge at an underlying of 1500 so as to be short 0.8 units of the S&P, and then subsequently the stock market declines again to 1400. Now the option is ATM again with a delta of 0.5, and we need to buy back 0.3 units of the underlying index in order to adjust our hedge ratio to the change in delta. We are back to where we started. The S&P index is again at 1400, we have a long ATM call (now with less time value, however), and we have a hedge versus the underlying by being short 0.5 units of the index. However, simply by adjusting our hedge ratio to the moves in delta as the market went up and down, we have sold an additional 0.3 of the underlying at 1500 (when delta went from 0.5 to 0.8) and bought back 0.3 of the underlying at 1400 (when the delta went from 0.8 to 0.5). Thus, we made a profit through our hedging operations, from selling the underlying stock market high and buying it back low. This is the same as saying that we were underhedged in an upward move and overhedged in a downward move of the market. At the end of the round trip, we have therefore pocketed a profit from adjusting the hedge ratio to the changes of the delta of the option (and lost some of its time value).

In general terms, the hedge ratio for an option changes with the price of the underlying. Ensuring that the hedge ratio is always equal to the delta of the option is called *delta hedging* of an option and requires continuous adjustment of the hedge ratio to changes in delta as the underlying market moves up and down.

Since delta changes when the underlying moves, our delta hedge will most always be a bit too much or a bit too little, such that market moves will result in a profit or loss from delta hedging. The key point is that the holder of a long option position will always profit from delta hedging, while the holder of a short option position will always lose.²

To illustrate that point, let's consider the example above from the perspective of an investor who is short a call on the S&P500 with a strike of 1400. In order to hedge, he needs to buy 0.5 of the index initially (at 1400). When the index increases to 1500 (delta increases from 0.5 to 0.8), he needs to buy an additional 0.3 of the index at 1500, and after the index returns to 1400, he needs to sell 0.3 again at 1400. Thus, as a mirror image of the example of a long option holder above, the short option holder is forced to buy high and sell low. Or, in other words, delta hedging a short option forces a trader into the unfavorable position of being underhedged in a declining market and overhedged in an increasing one.

In theoretical terms, the option price is the cost of the dynamic self-financing portfolio that will replicate the payoff of the option, including the

²At this point, we're assuming that there are no changes in other variables, such as the implied volatility of the option.

costs or benefits of delta hedging. And it's this sense in which Black-Scholes is said to be a no-arbitrage model.

From a trading perspective, exact dynamic replication of an option is impossible, so we can think of *the option premium as the market price for the right to be on the profitable side of delta hedging* (i.e. buying the underlying low and selling it high, and being underhedged in rising markets and overhedged in falling markets). As delta hedging means that the option is always hedged versus the underlying, there will be no profit and loss at expiry of the option. If the option expires ITM, it will be hedged with one underlying; if it expires OTM, it will be hedged with no underlying. In fact, delta hedging can be thought of as increasing the hedge ratio toward one as the option moves ITM and decreasing the hedge ratio toward zero as the option moves OTM, just as it is required for the option to yield no P&L at expiry. Hence, *delta hedging an option transfers all P&L of the option position from the expiry (difference of underlying versus strike) to the P&L of the delta hedging operations before expiry*. As a consequence, the market price of an option should depend on the cost of delta hedging that option. This was the key insight of Black and Scholes, which enabled them to determine the option price by the cost of a dynamically hedged, self-financing portfolio. While we shall not go into the mathematics and quantifications, we highlight some important consequences of that approach in qualitative terms:

- The option price is determined by profit from the dynamically hedged, self-financing portfolio that hedges a long position in the option. The profit from delta hedging a long option position increases with the volatility of the underlying market moves. In the example above, if the market had moved twice from 1400 to 1500 and back (in the same time period), we would have made twice the money from delta hedging (selling at 1500 and buying back at 1400 0.3 of the S&P500 index). And if the market had moved from 1400 to 1600 and back (with a delta of, say, 0.9 at 1600), we could have sold 0.4 at 1600 and bought 0.4 back at 1400. Thus, the higher the volatility (number and scale of moves in the underlying market) until expiry, the higher the profit from delta hedging a long option position.
- From a trading perspective, the option price is determined by the expected profit from delta hedging, which in turn is determined by the anticipated volatility in the underlying market until the expiry of the option. Consequently, *the option price reflects the anticipated volatility in the underlying market until expiry*. In other words, the option market reflects the market consensus about anticipated future volatility. If overall market participants anticipate a period of high volatility ahead, they see considerable potential to profit from delta hedging long option positions and should

thus be willing to pay a high price for the right to be on the profitable side of delta hedging (i.e. they'll pay a high option premium).

- The anticipated future volatility in the underlying market reflected in the option price is called the *implied* volatility of that option. And as we have seen above, in the framework of delta hedging (i.e. Black–Scholes option pricing theory), implied volatility is the main determinant of option premiums. Put otherwise, for a specific option, with given underlying, strike, and expiry, etc., its price depends only on the implied volatility. Therefore, instead of quoting the premium of an option in terms of dollars and cents, it can be and often is quoted as well in terms of the implied volatility associated with that premium.
- While the *implied* volatility is the *anticipated future* volatility of the market, the *actual* volatility occurring in the market until the expiry of the option is called the *realized* volatility. The implied volatility determines the premium of the option, whereas the realized volatility determines the actual P&L from delta hedging the option until expiry. Thus, *the P&L of delta hedging an option is the difference between the implied volatility and the realized volatility*. Put otherwise, the implied volatility reflects the market consensus at the time of purchase about expected future volatility, which determines the price of the option. In contrast, the realized volatility indicates the volatility that actually has occurred prior to the expiry of the option, which determines the P&L realized by delta hedging the option. If implied and realized volatility turn out to be the same (i.e. the market consensus about anticipated future volatility is matched by the actual volatility), then the premium of the option and the P&L from delta hedging it would be equal. If, on the other hand, the realized volatility turns out to be less than the anticipated volatility, the option premium is higher than the P&L from delta hedging, and therefore the one who bought the option (and delta hedged it) loses and the one who sold the option (and delta hedged it) wins.³

³Note that the calculation of delta should be based on the anticipated future volatility. Consider two traders who anticipate that future realized volatility will be 20%, while the current implied volatility is 30%. Both are motivated to gamma trade the difference between the implied volatility and the volatility they anticipate (i.e. to sell the option and delta hedge it). Trader A calculates all his hedge ratios using a volatility of 20% (i.e. the volatility he anticipates), while Trader B calculates all his hedge ratios using a volatility of 30% (i.e. the volatility currently implied by the pricing of the option). Let's say that the realized volatility actually does turn out to be 20%, as these two traders anticipated. While Trader A will have realized a profit in line with the Black–Scholes model, the P&L of Trader B, using a number for delta which is different from the Black–Scholes model, can be different (higher or lower than the profit of trader A).

- Therefore, trading in options and delta hedging them means taking views on the spread between implied and realized volatility. We buy options and delta hedge them if we believe that the realized volatility of the underlying will be greater than their implied volatility. Since option prices depend on the volatility anticipated by the market, their actual ending value at expiry can only be known after expiry, when the realized volatility and thereby the P&L from delta hedging are known. Before expiry, trading in options always involves assessing the market consensus about future anticipated volatility as reflected in the implied volatility of the option prices versus our own view about future volatility.
- The option value today is *not* determined by the price of the underlying at expiry. In the Black–Scholes approach, the option value today depends on the volatility of the price path of the underlying. Put in simple terms, for pricing options it does not matter *at which level* the underlying of the option will end, only *how* it gets there: in a volatile or a smooth fashion.

Note how a simple analysis of the delta of an option has naturally translated into a hedging (delta hedging), pricing (Black–Scholes), and trading (implied versus realized volatility) strategy.

Theta, Vega, and Gamma

Similarly to delta, which assesses the price of an option as a function of the underlying security, one can also analyze the way the option price would change in relation to changes in other determining variables. In particular:

- Theta is defined as $\frac{\partial \text{PriceOption}}{\partial \text{Time}}$ and indicates the changes in the value of the option as a function of time.
- Vega⁴ is defined as $\frac{\partial \text{PriceOption}}{\partial \text{ImpliedVolatility}}$ and shows the change in the value of the option as a function of changes in implied volatility.
- Gamma is defined as $\frac{\partial \text{Delta}}{\partial \text{PriceUnderlying}}$ and quantifies the change in delta as a function of the change in the price of the underlying (i.e. it's the second derivative of the option price with respect to the price of the underlying, i.e. the slope of the line in Figure 19.1). Thus, gamma is close to zero for far ITM and far OTM options and maximal for ATM options.

Given a certain realized volatility of the market, the P&L from delta hedging will be higher, if the (given) price changes of the underlying induce a

⁴For both systematic and philological reasons, we would like to suggest replacing “vega” with “sigma,” but stick in the following to the conventional term.

higher change in delta (and thereby require a larger amount of the underlying to be bought or sold in order to maintain the delta hedge). Thus, the higher the gamma of an option, the larger the changes in delta and the larger the P&L from delta hedging, given a certain realized volatility. As a consequence, the trading strategy developed above of delta hedging an option and thereby exploiting the difference between implied and realized volatility requires options with a sufficiently large gamma. Both far ITM and far OTM options are not suitable for that strategy. And also ATM options which have a low gamma for different reasons, in particular because of a long time to expiry, require another trading strategy than delta hedging. Thus, this brief and purely qualitative review of option pricing yielded a hedging and trading strategy (delta hedging) for options with high gamma as well as a segmentation of the option market into those options for which this approach is suitable and into those for which it is not. Subsequently, we shall build on this result so as to classify the option market into different segments and to associate them with their respective suitable trading strategies.

CLASSIFICATION OF OPTION TRADES

Relationship between Gamma and Theta

Theta measures the time decay of the option. Figure 19.3 shows schematically that theta becomes increasingly negative as the option moves closer

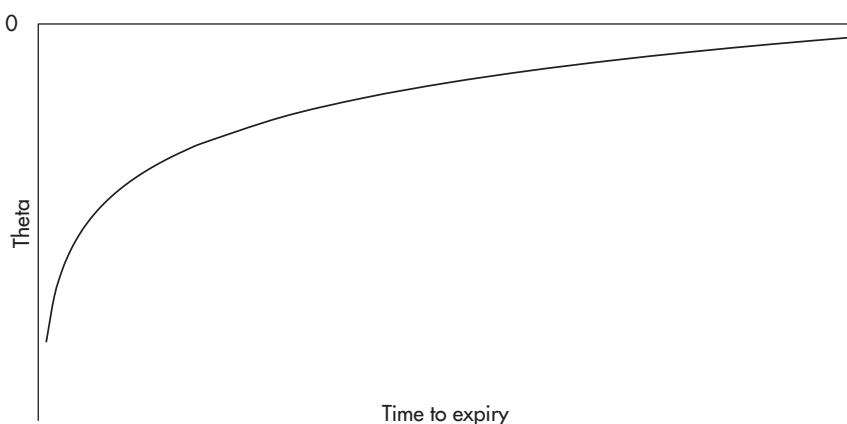


FIGURE 19.3 Theta of an ATM option as a function of the time to expiry (schematic).
Source: Authors.

to expiry. That is, options with a long time to expiry have a small negative theta, while options with a short time to expiry have a high negative theta (all else being equal).⁵

It can be deduced from the Black–Scholes formula that (all else being equal) a decrease of theta (i.e. theta becoming more negative) is linked to an increase of gamma. More precisely, an increase in gamma by 1 is equivalent to a decrease in theta by $\frac{2}{P^2\sigma^2}$ where P is the price of the underlying and σ is the implied volatility (again, with all other parameters assumed to remain unchanged).

Together, this results in options with a long time to expiry having a relatively high theta (small negative number), thus a relatively small gamma (small positive number). As the time to expiry decreases (and assuming that the ATM option remains ATM), theta decreases, from a number closer to zero to a larger negative number, while gamma increases, from a number closer to zero to a larger positive number. Consequently, the exposure of an ATM option to changes in delta becomes increasingly large as the expiry date approaches. This is depicted in Figure 19.4, which illustrates that the move of delta from 0 to 1 becomes more and more concentrated around the ATM point as the time to expiry decreases.

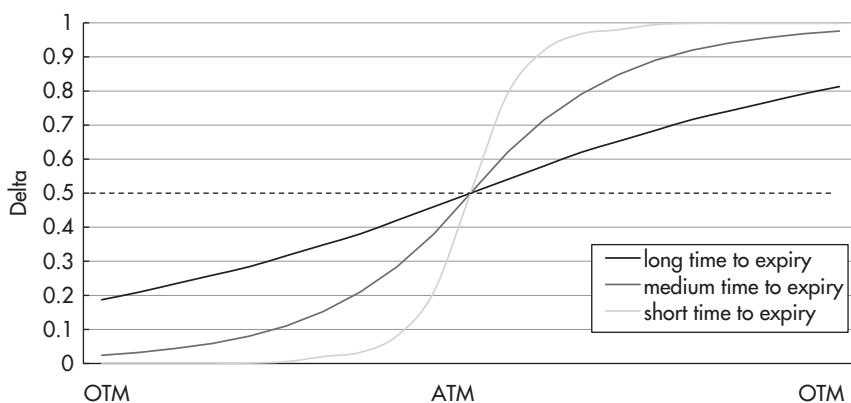


FIGURE 19.4 Delta of an option as a function of the difference between the price of the underlying and the strike price, as the time to expiry of the option becomes shorter and shorter.

Source: Authors.

⁵While it is possible for theta to be positive for ITM options, the chart depicts the usual situation for ATM options.

This mathematical result also makes sense intuitively: using the example above, imagine we had a 10-year (10Y), rather than a 3M, call to buy the S&P500 index at 5000. Given the very long time to expiry, that option will have some (time) value, which decreases every day only by a small amount (low negative theta), and will hardly change if the S&P index moves from 1400 to 1500 (low positive gamma, thus a flat delta curve in Figure 19.4). On the other side of the extreme, consider an ATM option that expires tomorrow. Like any ATM option, it has a delta of 0.5, but at expiry tomorrow it will either be ITM with a delta of 1 or OTM with a delta of 0. Tomorrow, it will have lost all time value (theta approaching negative infinity), and the slightest move of the index up or down will decide between a delta of 1 or 0. As illustrated in Figure 19.4, when the time to expiry decreases, the gamma of an ATM call option increases (approaching positive infinity). That is, the delta curve becomes steeper and steeper until at expiry it is a binary curve with a value of 0 below the strike and a value of 1 above the strike.

Segmentation of the Volatility Surface

Turning from option theory, from which we have picked only those parts which are needed to construct the right trading strategies, to its practical application, we can summarize the result as follows:

- *The prices of ATM options with short times to expiry mainly depend on price moves in the underlying security.* Due to the high gamma of options with short times to expiry, delta changes significantly when the underlying moves. This means that price moves in the underlying security have a large impact on the P&L from delta hedging an option and therefore on its premium as well.
- *The price of ATM options with long times to expiry mainly depend on moves in the implied volatility.* In contrast to options with short times to expiry, for a low gamma option a move in the underlying does not result in a meaningful change of delta and thus has no significant impact on the P&L from delta hedging and therefore nor on the option price. In the limit of very long expiries, the slope of the delta line in Figure 19.4 approaches 0, which means that moves in the underlying security do not result in a change in the delta, and therefore they have no impact on the P&L from delta hedging and therefore neither on the option price. On the other hand, given the long time value of the option, even a small change in the implied volatility has a significant effect on the option price. This intuition is reflected in the mathematical fact that (excluding extremely

long times to expiry) vega rises with increasing time to expiry,⁶ while gamma declines.

The impact of delta hedging is a decreasing function of the time to expiry. For our practical approach this means that delta hedging is an appropriate trading strategy only for options with a short time to expiry. For swaptions, this usually covers a range of expiry times up until six months or, in some cases, one year. Swaptions with times to expiry of three years or more have values that are largely independent of changes in the value of the underlying and mainly influenced by changes in the implied volatility.

Thus, the swaption volatility surface (i.e. the two-dimensional grid of swaptions sorted by time to expiry of the option and by time to maturity of the underlying swap) should be divided into a gamma sector, where delta hedging is the appropriate trading strategy, and into a vega sector, where it is not (Figure 19.5). Usually, options with a time to expiry of roughly two years fall in a gray area between those two sectors, and the appropriate trading strategy must be determined on a case-by-case basis. The precise location of the border between the gamma and vega sectors differs for each option market, and often the gray area is somewhat tilted to the right (see Figure 19.20, where we provide more details for the case of JPY swaptions).

This segmentation of the volatility surface into a gamma and vega sector provides the most basic classification of option trades:

- Options with a short expiry horizon provide a strong link to the underlying and thus an exposure to changes in the realized volatility (i.e. the price moves of the underlying).
- Options with a long expiry horizon have less contact with the underlying and are therefore mainly exposed to changes in the implied volatility.

And consequently:

- Options in the gamma sector can be appropriately traded in the Black-Scholes framework of delta hedging.

⁶Strictly speaking, vega depends on the time to expiry, t , in accordance with the term $\frac{\sqrt{t}}{e^t}$. As a result, vega first increases with t . For higher values of t , on the other hand, the exponential function increases at a faster rate than does the square root function and vega decreases with t . The specific point at which vega shifts from being an increasing function of time to a decreasing function of time also depends on additional parameters, including the implied volatility. However, for common values of implied volatility and times to expiry, it is relatively safe to assume that vega increases with time.

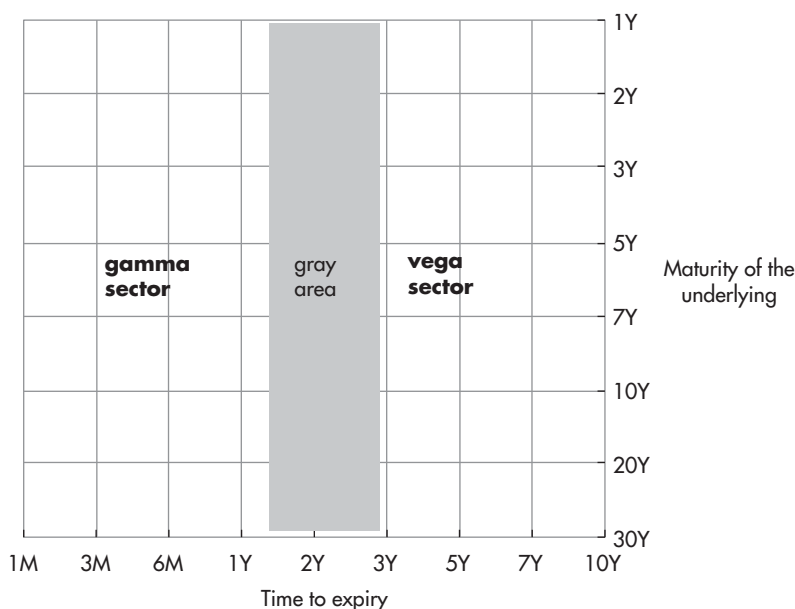


FIGURE 19.5 Classification of the volatility surface into a sector suitable for gamma trades and another suitable for vega trades.

Source: Authors.

- Options in the vega sector need an analysis and trading strategy different from delta hedging. We shall show below that a statistical analysis of the mechanisms of the implied volatility can provide appropriate trading strategies for the vega sector. For example, a principal component analysis (PCA) is able to reveal the way implied volatility affects the volatility surface and how to exploit this knowledge. Hence, we shall not attempt to build an option pricing model for the vega sector, which is superior to Black–Scholes, but rather use statistical techniques to find value within that sector.

We restrict our discussion to the volatility surface of at-the-moneyforward (ATMF) options (i.e. those with the strike of the swaption at the forward rate of the underlying swap). Including different strikes would expand the volatility surface to a volatility cube, with the additional dimension representing the skew. The discussion of skew in the gamma sector requires an option model,

while in the vega sector it can be addressed through the same statistical methods (e.g. via expanding the dimensionality of the PCA).⁷

Classification of Option Trades in the Gamma Sector

Continuing with our classification within the gamma sector, one is free to choose the *type* of exposure to the realized volatility one wants:

- ① Without delta hedging, one is exposed to the overall realized volatility of the underlying until expiry of the option. This corresponds to choosing exposure to the payoff profile of an option at expiry. We refer to this strategy as option trade type ①.
- ② With delta hedging, one is exposed to the continuous realized volatility of the underlying until expiry of the option. This corresponds to choosing exposure to P&L from continuous adjustments to the hedge ratio (in the Black–Scholes framework). We refer to this strategy as option trade type ②.

While theoretically speaking, ① and ② only differ in the number of time periods, the exposure in market and trading terms is different:

- ① gives a *conditional* exposure to the overall move of the underlying, while being largely unaffected by continuous realized volatility;
- ② gives an exposure to the realized volatility of the underlying until expiry, while being largely unaffected by the overall move of the underlying and its price at expiry.

For these two types of option trades in the gamma sector, implied volatility only matters at the beginning, as it determines the price to enter into these strategies. Afterward, their performance only depends on realized

⁷Indeed, a “skew factor” can be identified when running a PCA on the volatility cube in the vega sector. However, given limited liquidity in many of the OTM series, the input data often represent not actual price moves but constructed data. Correspondingly, the output of a PCA skew factor does not reveal the mechanisms of real market action but rather the assumptions of the model a trader used to construct the artificial data series. Thus, expanding the reach of a PCA from the volatility surface to the volatility cube will only be meaningful when liquidity across the volatility cube has increased and provides real market data as input.